

Calculus BC

Midterm 2 Practice Set

16 Required Problems, 1 bonus problem, 90 Mins

Try not to use a calculator unless you really have to

1. $\int 3x^2 - 2x + 3 \, dx$, given the point (1,5)

2. $\int \frac{1}{6-12x} \, dx$

3. $\int \frac{d}{dx}(\csc x)$

4. $\int_0^3 |2x - 1| \, dx$

5. $\frac{d}{dx} \int_0^{2x^2} x^2 + 4 \, dx$

6. $\int_0^{\frac{\pi}{6}} \frac{\cos \theta}{1+2\sin \theta} \, d\theta$

7. Find the total area between the x-axis and the curve $f(x) = 2x^2 - 8$ from $x = 0$ and $x = 3$

8. Given $\frac{d^2y}{dx^2} = 6x - 4$, $y = 4$, $y' = 5$, and $x = 2$, find the original equation

9. Find the value of c that satisfies the mean value theorem $f(x) = x^3 - x^2 - 2x$ in the interval $x = [-1, 1]$

10. Find all the x values of the points of inflection in the function $f(x) = -x^4 + 4x^3 - 4x + 1$

11. Use differentials to approximate the value of $\frac{1}{\sqrt{50}}$

12. Given $\frac{dy}{dx} = x^2 - \frac{1}{2}y$ and $y(0) = 2$, use Euler's method starting at $x = 0$ with two steps of equal size to approximate $y(1)$

13. A poster is to have a total area of 180in^2 with 1-inch margins at the bottom and sides, and a 2-inch margin at the top. What dimensions of the print would give the largest printed area?

14. A scalene triangle has an angle θ , with two adjacent sides a and b . Given that at a certain instant, $a = 8\text{cm}$, $b = 12\text{cm}$, $\theta = \frac{\pi}{4}$ and $\frac{d\theta}{dt} = 0.02\text{ rad/s}$, how fast is the area of the triangle changing?

15. The temperature of a park in Celsius in a 4-hour period is modeled by $f(x) = 10 + 5 \sin\left(\frac{\pi x}{4}\right)$. Using the integral mean value theorem, find the average temperature at the park.

*Assume the beginning of the period is $x = 0$ and the end is $x = 4$.

16. A swimming pool has a width of 30 feet and a length of 75 feet. The depth of the pool at certain points is represented in the table below.

Length (ft)	0	25	50	75
Depth (ft)	3	3	5	7

Using the table above and the right rectangular approximation method, estimate the volume of the swimming pool.

17. Bonus Question (This may be an extra credit problem in the midterm; it was like that last year). Use summation to find the area between the function $f(x) = x^2 - 6x + 8$ and the x-axis in the interval $x = [0, 2]$. You may use integrals and the fundamental theorem of calculus to check your answer afterwards.

- $\sum_{k=1}^n c = nc$
- $\sum_{k=1}^n k = \frac{n(n+1)}{2}$
- $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$
- $\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$

Key:

1. $f(x) = x^3 - x^2 + 3x + 2$

2. $-\frac{1}{12}\ln|6 - 12x| + C$

3. $\csc x + C$

4. 6.5

5. $4x^3 + 16$

6. $\ln \sqrt{2}$

7. $\frac{46}{3}$

8. $y = x^3 - 2x^2 + x + 2$

9. $c = -\frac{1}{3}$

10. $x=0, x=2$

11. $\frac{99}{686}$

12. 1.25

13. $2\sqrt{30} - 2 \ln x - 3\sqrt{30} - 3 \ln$, surely, correct me if I'm wrong because I'm horrible at optimization

14. $\frac{12\sqrt{2}}{25} \text{ cm}^2/\text{second}$, surely

15. 10 degrees celcius

16. 5250 cubic ft

17. $\frac{20}{3}$